

Classification and Prediction

***Unveiling Patterns and Making Predictions from
Data***

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INTRODUCTION TO CLASSIFICATION & PREDICTION

WHAT IS CLASSIFICATION?

- A SUPERVISED LEARNING TASK.
- GOAL: TO ASSIGN NEW, UNSEEN DATA TO PREDEFINED CATEGORIES OR CLASSES.
- THE MODEL LEARNS FROM A LABELED TRAINING DATASET WHERE THE CORRECT CLASSES ARE ALREADY KNOWN.

WHAT IS PREDICTION?

- SIMILAR TO CLASSIFICATION, BUT PREDICTS A CONTINUOUS VALUE INSTEAD OF A CLASS.
- THE MOST COMMON FORM IS REGRESSION ANALYSIS.

REAL-WORLD EXAMPLES:

- CLASSIFICATION: EMAIL SPAM DETECTION (SPAM VS. NOT SPAM), MEDICAL DIAGNOSIS (HAS DISEASE VS. DOES NOT HAVE DISEASE).
- PREDICTION: FORECASTING HOUSE PRICES BASED ON FEATURES LIKE SIZE AND LOCATION, PREDICTING FUTURE SALES FIGURES.

Decision Tree Induction

Key Algorithms:

- **ID3 (Iterative Dichotomiser 3):** Uses Information Gain to select the best attribute for each split. It aims to reduce uncertainty (entropy) the most.
- **CART (Classification and Regression Trees):** Uses the Gini Index to measure impurity. It creates binary (two-way) splits and works for both classification and regression.

Tree Pruning: A vital step to prevent overfitting, where the tree becomes too complex. Pruning simplifies the tree to improve its performance on new, unseen data.



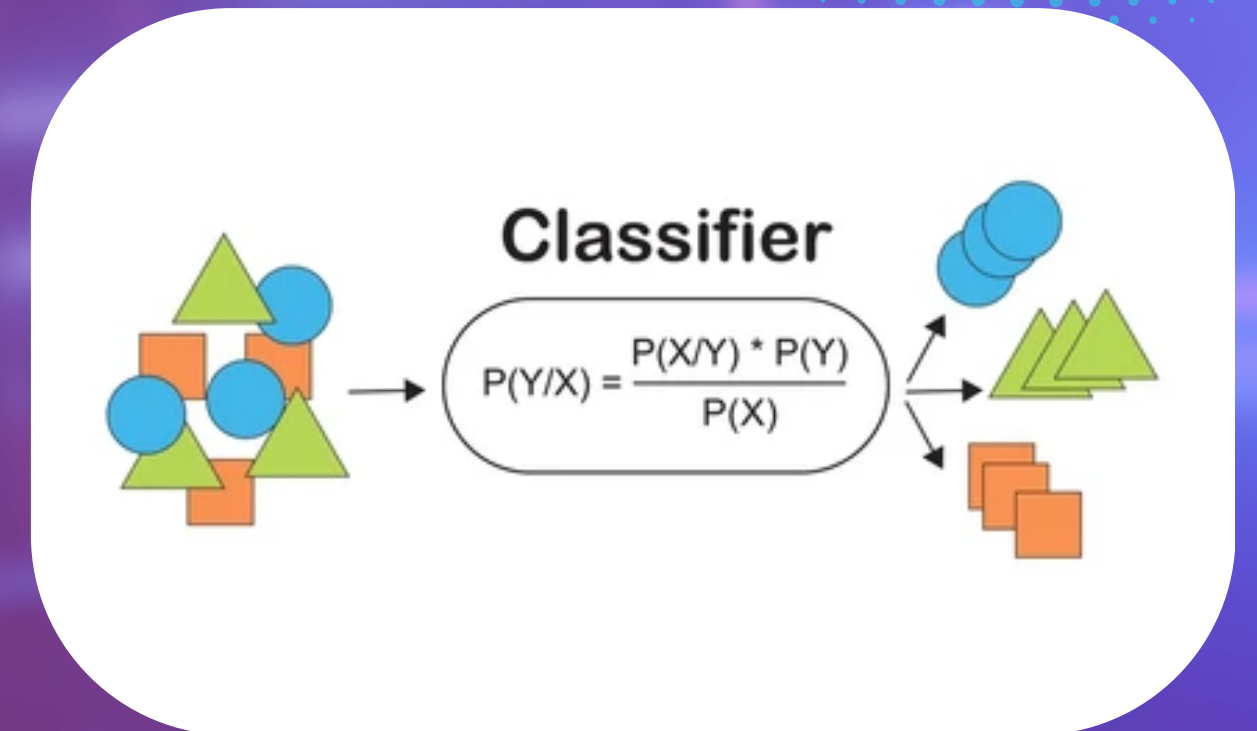
Bayesian Classification Methods

Based on Probability: Bayes' Theorem

- A theorem that calculates the probability of an event happening based on prior knowledge of conditions related to that event.
- Formula: $P(\text{Class} | \text{Data}) = (P(\text{Data} | \text{Class}) * P(\text{Class})) / P(\text{Data})$

Naive Bayesian Classifier

- A simple yet powerful probabilistic classifier.
- The "Naive" Assumption: It assumes that all features are independent of one another, given the class.
- How it Works: It calculates the probability of a data point belonging to each possible class and assigns it to the class with the highest probability.



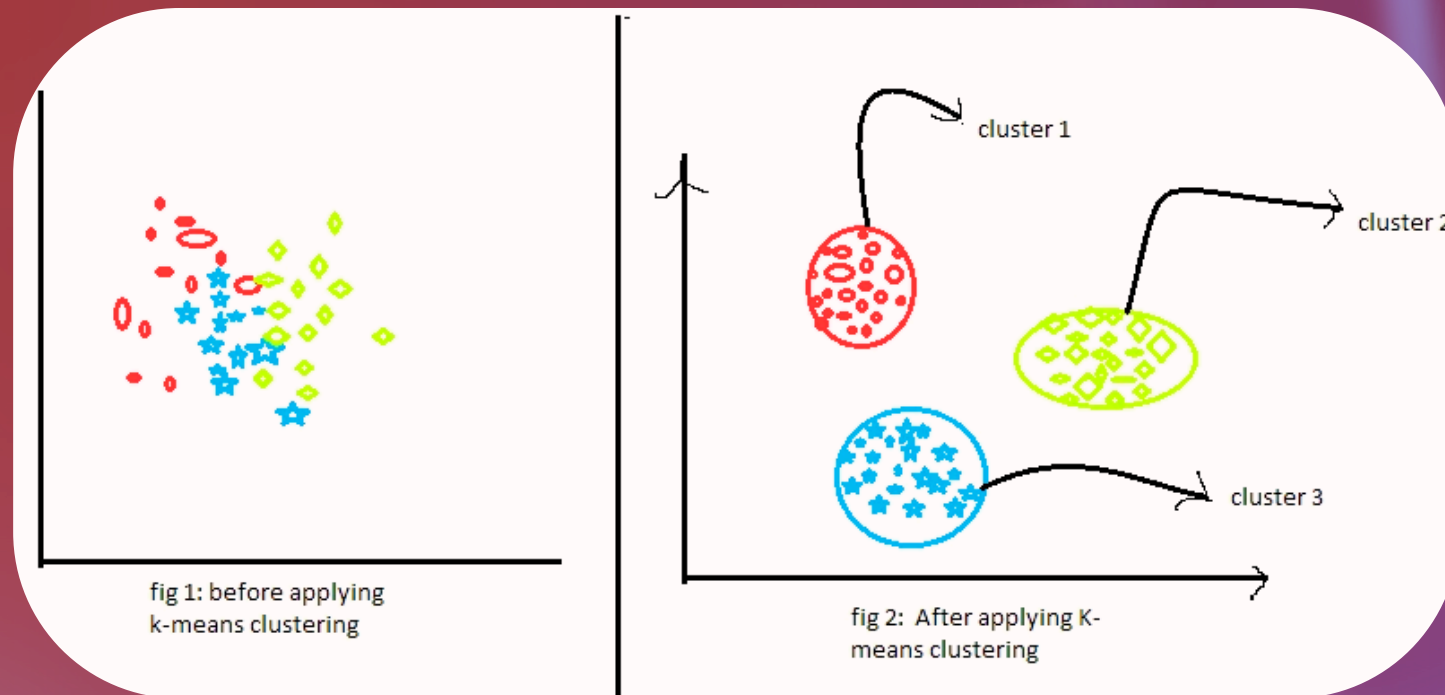
Advanced Classifiers: Backpropagation & SVM

Support Vector Machines (SVM)

- Finds the optimal hyperplane (a decision boundary) that best separates data points into classes.
- Goal: To maximize the margin—the distance between the hyperplane and the nearest data points of any class.
- Key Advantage: Very effective in high-dimensional spaces and can handle complex, non-linear relationships using a technique called the "kernel trick."

Backpropagation

- The primary algorithm for training Artificial Neural Networks (ANNs).
- Process:
 - The network makes a prediction.
 - The error between the prediction and the actual value is calculated.
 - This error is propagated backward through the network's layers.
 - The connection weights between neurons are adjusted to minimize the error.



Lazy Learners: K-Nearest Neighbors (KNN)

- An instance-based learning algorithm. It doesn't build a model.
- Called a "Lazy Learner" because it does no work during the training phase. It simply stores the entire training dataset.

How KNN Classifies Data:

- When a new data point arrives, it finds the 'k' closest data points (neighbors) to it from the training set.
- It looks at the classes of those neighbors.
- The new data point is assigned to the class that is most common among its 'k' neighbors (a majority vote).

Key Considerations:

- Choosing the right value for 'k' is critical.
- Can be computationally slow during classification if the dataset is large.

Evaluating and Improving Classifier Performance

Key Metrics for Evaluation:

- Accuracy: The percentage of correct predictions. (Simple, but can be misleading for imbalanced datasets).
- Confusion Matrix: A table that breaks down performance into:
 - True Positives (Correctly identified)
 - True Negatives (Correctly rejected)
 - False Positives (Incorrectly identified)
 - False Negatives (Incorrectly rejected)
- Precision & Recall: More nuanced metrics that measure correctness and completeness.

Techniques to Improve Accuracy:

- Cross-Validation: Repeatedly splitting the data to get a more reliable performance estimate.
- Ensemble Methods (Bagging, Boosting): Combining multiple models to create a more powerful "super model" that is more accurate than any single one.



Thank You

